

System Design Basics

Lesson 2: Scalability and Load Balancing

Scalability and Load Balancing - Study Notes

Key Concepts

- Horizontal vs. vertical scaling
- Stateless vs. stateful services
- Load balancer types

Summary

Scalability ensures a system can handle increased load. Horizontal scaling adds more instances, while vertical scaling upgrades hardware. Stateless services are easier to scale because any instance can serve any request. Load balancers distribute traffic across instances, improving availability and fault tolerance. Techniques like caching and sharding further enhance performance.

Important Points

- Statelessness simplifies scaling.
- Load balancers can be round robin, least connections, or IP hash.
- Monitoring latency helps identify bottlenecks.

Example

Use Nginx as a reverse proxy to balance requests between multiple app servers, adding a Redis cache for session data.

Quick Review

- How does statelessness aid scaling?
- Name two common load balancing algorithms.

